Introduction to Data Assimilation
Lecture 5 Supplementary

Quanling Deng*

March 26, 2026

1 Filtering Stability for the 1D Kalman Filter

1.1 1D Kalman Filter

Recall that the forecast and observational models are below.

$$\text{Forecast:} \quad u_{m+1} = F\, u_m + \sigma_{m+1}, \qquad \sigma_{m+1} \sim \mathcal{N}(0, r). \tag{1}$$
$$\text{Observation:} \quad v_{m+1} = g\, u_{m+1} + \sigma^o_{m+1}, \qquad \sigma^o_{m+1} \sim \mathcal{N}(0, r^o). \tag{2}$$

The Kalman Filter algorithm is: Given $(\bar{u}_{m|m}, r_{m|m})$:

$$\text{Forecast:} \quad \bar{u}_{m+1|m} = F\, \bar{u}_{m|m}, \qquad r_{m+1|m} = F r_{m|m} F^* + r. \tag{3}$$
$$\text{Analysis:} \quad K_{m+1} = \frac{g\, r_{m+1|m}}{r^o + g^2 r_{m+1|m}}, \tag{4}$$
$$\bar{u}_{m+1|m+1} = \bar{u}_{m+1|m} + K_{m+1}\big(v_{m+1} - g\, \bar{u}_{m+1|m}\big), \tag{5}$$
$$r_{m+1|m+1} = (1 - K_{m+1} g)\, r_{m+1|m}. \tag{6}$$

1.2 Mean-square stability and the scalar Riccati map

Define the posterior variance $P_m := r_{m|m}$ and the prior $S_m := r_{m+1|m} = |F|^2 P_m + r$. The scalar Riccati recursion is

$$P_{m+1} = \Phi(S_m) := \frac{S_m\, r^o}{r^o + |g|^2 S_m}, \qquad S_m = |F|^2 P_m + r. \tag{7}$$

The map Φ is increasing and concave on $[0, \infty)$ with $\Phi(0) = 0$ and $\Phi(S) \nearrow r^o/|g|^2$ as $S \to \infty$.

Definition 1.1 (Mean-square stability)**.** *The filter is* mean-square stable *if P_m remains bounded and $P_m \to P_\infty$ as $m \to \infty$ for some finite $P_\infty \geq 0$.*

Proposition 1.2 (Existence/uniqueness of P_∞)**.** *If $g \neq 0$ and $r, r^o > 0$, the coupled map $P \mapsto \Phi(|F|^2 P + r)$ has a unique fixed point $P_\infty \in (0, r^o/|g|^2)$, and $P_m \to P_\infty$ from any $P_0 \geq 0$.*

*Yau Mathematical Science Center, Tsinghua University, Beijing, China 100048. E-mail address: qldeng@tsinghua.edu.cn

1.3 Closed-form steady state and closed-loop factor

Let $a := F$. Eliminating $S = |a|^2 P + r$ in $P = \Phi(S)$ gives the quadratic

$$|g|^2|a|^2P^2 + \big(|g|^2r + r^o(1-|a|^2)\big)P - r\,r^o \;=\; 0. \tag{8}$$

The positive root is

$$P_\infty \;=\; \frac{-B + \sqrt{B^2 + 4\,|g|^2|a|^2 r\,r^o}}{2\,|g|^2|a|^2}, \qquad B := |g|^2 r + r^o(1-|a|^2). \tag{9}$$

At steady state the prior is $S_\infty = |a|^2 P_\infty + r$ and the gain

$$K_\infty = \frac{S_\infty g^*}{r^o + |g|^2 S_\infty}, \qquad 1 - K_\infty g = \frac{r^o}{r^o + |g|^2 S_\infty} \in (0,1).$$

The *closed-loop factor* for the mean ($u_{m+1|m+1} = (1-Kg)u_{m+1|m} + Kv_{m+1}$) is

$$\alpha_\infty := a(1-K_\infty g), \qquad |\alpha_\infty| = |a|\,\frac{r^o}{r^o + |g|^2 S_\infty} < 1,$$

so the mean recursion is stable.

Remark 1.3 (GPS vs. inertial guidance). Inertial sensors (model) drift; periodic GPS fixes (observations) pull the solution back. Even if the inertial prediction blows up, sufficiently frequent/accurate GPS updates yield a stable fused estimate.